

(TWR) Thrust-to-weight Ratio Hand calc

$$\text{Max thrust} = 4 \text{ kg}$$

$$W_{\text{max}} = \text{Max weight}$$

$$(TWR) = 2:1 = \frac{2}{1} = 2 \text{ kg}$$

$$\frac{4 \text{ kg}}{W_{\text{max}}} = \frac{2 \text{ kg}}{1 \text{ kg}}$$

$$W_{\text{max}} = 2 \text{ kg}$$

$$W_{\text{Dmax}} = \text{Max drone arm weight}$$

$$W_{\text{comp}} = 1 \text{ kg} \quad \text{drone components}$$

$$W_{\text{min}} = 0.5 \text{ kg} \quad \text{min payload}$$

$$W_{\text{Dmax}} = \frac{(W_{\text{max}} - W_{\text{comp}} - W_{\text{min}})}{4}$$

4

← To account for quadcopter structure

$$W_{\text{Dmax}} = 0.125 \text{ kg}$$

One drone arm has to be less than or equal to 0.125 kg or 125 g. to be useable.